

Forecasting Amazon (AMZN) Stock Prices Using Time Series Analysis

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1. Introduction and Objective

Financial research together with professional practice heavily depends on stock market prediction analysis as a fundamental subject. Accurate stock market value estimation helps investors develop superior investment approaches and track their investment portfolio risks. Precise financial prediction remains difficult to execute because of the timewise complexities of series along with their unpredictable fluctuations and random interference. The research creates prediction models to study Amazon (AMZN) stock closing prices recorded daily throughout previous periods. A central aim of this research intends to: The forecasting ability of stock market values should be measured through implementation of traditional forecasting approaches specifically ARMA. The analysis implements LSTM networks for deep learning model implementation to evaluate their forecasting accuracy against conventional methods. The project explores GAN-generated synthetic time series data for enhancing model performance assessment.

2. Data Collection and Preprocessing

Dataset Description

Daily trading information covering a ten-year period was retrieved from Yahoo Finance starting from January 1, 2010 until December 31, 2020. The dataset included six variables consisting of Open, High, Low, Close, Adjusted Close and Volume. The Close price function served as the forecasting target because stock market analysts commonly use day-end transaction prices. The long historical dataset enables exceptional modeling possibilities both in statistics and machine learning techniques.

Preprocessing Steps

Financial data in time series format requires specialized data processing to become ready for modeling purposes. The following steps were taken:

- **Handling Missing Values:**
The procedure for handling days when no trading occurred involved filling forward the blank values.
- **Stationarity Transformation:**
The combination of first-order differencing and logarithmic transformation made the time series more stable both in terms of its tendency and its variance before model application.
- **Normalization:**
The data received MinMaxScaler normalization to reach better results with neural network models.
- **Data Splitting:**
Training used 80 percent of the total data while the remaining 20 percent served for testing purposes. Testing of forecasting models used the remaining 20% of the data.

3. Time Series Modeling and Diagnostics

Forecasting models were divided into two main categories:

Traditional statistical models and modern deep learning models.

3.1 ARMA Model

The ARMA (AutoRegressive Moving Average) model contains elements from both autoregression along with moving average forecasting.

Future predictions rely on two components which include autoregression (AR) for value forecasting from historical data and moving average (MA) techniques that use past prediction errors.

Moving Average (MA) utilizes previous forecast errors to predict future values.

Model fitting process:

- The research team conducted a search through multiple combinations of $AR(p)$ and $MA(q)$ values.
- The selection of the best model occurred through minimizing the Akaike Information Criterion (AIC).
- Tests of residual diagnostics showed that errors followed the characteristics of white noise.

3.2 LSTM Model

The Long Short-Term Memory networks (LSTMs) represent advanced Recurrent Neural Networks (RNNs) made to recognize extended dependencies within time-based datasets.

Why LSTM?

- Stock prices are influenced not only by recent events but also by trends from weeks or months ago.
- These networks remember information through many time steps unlike traditional ARMA models which usually focus on short-term prediction.

Model Architecture:

- Two LSTM layers, each with 50 units.
- Dropout regularization between layers.
- Dense output layer to produce the forecasted value.

Training:

- Trained for 10 epochs with a batch size of 32.
- Loss function: Mean Squared Error (MSE).
- Optimizer: Adam.

3.3 GAN-Augmented LSTM

GANs (Generative Adversarial Networks) create a new approach to time series forecasting through their production of synthesized realistic data sequences.

TimeGAN Architecture:

- The generator developed artificial stock price sequences as part of its operation.
- The discriminator operated to detect whether provided sequences were real or artificial.
- The training process continued until the generator succeeded in generating sequences that the discriminator could not differentiate from authentic ones.

Integration with LSTM:

LSTM models received training with data combining both authentic historical data and synthetic data from the system.

4. Forecasting and Evaluation

The test dataset performance evaluation used the following metrics:

Metric	Description
MAE (Mean Absolute Error)	Measures average magnitude of errors.
RMSE (Root Mean Squared Error)	Penalizes large errors more heavily.
R ² (Coefficient of Determination)	The R ² value demonstrates how accurately predicted values align with actual values where perfection .

Performance Results

Model	MSE	RMSE	MAE	R ²
		High		
			High	Very Low
		0.0055		
Random Walk	High	High	High	Low
ARMA	~0.0024	~0.049	~0.038	~0.82
LSTM	0.001620	0.040247	0.032085	0.955541
LSTM + GAN	Similar to LSTM	Slight Improvement	Slight Improvement	Slight Gain

The LSTM model delivered the greatest R² value of 95.5% which reflected an extremely strong correlation of predicted stock prices to actual market movements.



5. Discussion and Conclusion

Interpretation of Results

- **Traditional Models:**
Short-term market trends could be detected by ARMA models although these models faced difficulties when analyzing complex market fluctuations.
- **Deep Learning Models:**
LSTM networks used their ability to learn both short-term and long-term patterns in stock price movement effectively.
- **GAN-Augmented Learning:**
The addition of synthetic data through GAN improved the training set marginally yet made GAN training more unstable.

Challenges Encountered

- The models excluded external unpredicted factors that impact stock prices during their analysis.
- GANs needed precise adjustments for achieving stable training operations.
- The implementation of LSTM models required substantial time for both high computation demands and parameter adjustments.

Future Work

- The proposed model integration includes economic indicators as well as news sentiment scores as external features.
- The research should examine Transformer-based time series models including Informer and TimesNet.
- GAN training methods need improvement to generate synthetic data which better resembles the original data.